

1. Formulas vs. Functions

- **Formula:** A user-defined statement used to calculate values (e.g., =A1+A2+A3). It can perform basic mathematical operations like addition (+), subtraction (-), multiplication (*), and division (/).
- **Function:** A predefined formula already built into Excel (e.g., =AVERAGE()).
- **Syntax & Parts of a Function:** A function must always begin with an equal sign (=), followed by the specific function name, and one or more **arguments** enclosed in parentheses. Multiple arguments within a function must be separated by commas.

Common Built-In Functions

- **SUM:** Adds all the values in the cell argument range.
- **AVERAGE:** Calculates the sum of the cells and divides it by the total number of cells in the range to find the arithmetic mean.
- **COUNT:** Counts the number of cells that contain numerical data.
- **COUNTA:** Counts the number of records/cells that are not blank in a single-column table (includes empty text "").
- **COUNTIF:** Counts the number of cells in a range that meet a specific logical criterion or condition (e.g., =COUNTIF(C5:C14,">1500")).
- **MAX:** Determines the highest cell value in the range.
- **MIN:** Determines the lowest cell value in the range.

2. Cell and Range References

- **Cell Reference:** A combination of a column letter and a row number that identifies a specific cell (e.g., A1).
- **Range Reference:** A block of two or more cells separated by a colon (e.g., A1:C2 includes 6 cells from cell A1 through C2).
- **Absolute Reference:** Uses dollar signs to "lock" a row and/or column reference so it remains constant when a formula is copied to other cells (e.g., =\$E\$1).

Replicating Formulas with the Fill Handle

The **Fill Handle** is a small square located in the lower-right corner of a selected cell.

1. Select the cell containing the formula.
2. Hover over the corner until the cursor changes to a thick black cross.
3. Click, hold, and drag it down or across adjacent cells to instantly copy the formula and cell formatting.

3. Excel Formula Errors & Corrections

Error Code	Primary Cause	How to Fix It
#NAME?	Misspelling a function name (e.g., entering =su(A2,B2) instead of SUM).	Check the formula bar and ensure the function spelling is valid.
#DIV/0!	Attempting to divide a number by zero or an empty cell.	Use the IFERROR function to handle empty fields elegantly (e.g., =IFERROR(A2/B2,0)).
#REF!	A cell reference that the formula relies on was deleted or missing suddenly.	Before deleting cells, paste formulas as values, or use absolute references (\$) to lock cells.
#NULL!	Forgetting proper syntax separators (e.g., putting a space instead of a comma or colon between arguments).	Ensure proper syntax: use a comma (,) to separate independent arguments or a colon (:) for ranges.
#VALUE!	The formula contains non-numerical values or incorrect data types where numbers are expected.	Ensure all cells used in arithmetic operations contain numeric values, not text strings.
#####	Not an actual error; the column is just too narrow to display the formatted date or number.	Double-click the column boundary line at the top to auto-adjust and expand the column width.

4. Calculations: Percentages and Complex Formulas

Percentage Basics

- **Finding Percentage Form:** $=(\text{Value}/\text{Total})*100$.
- **Calculating a % of a Number:** To find a direct percentage, type the value multiplied by the percentage sign (e.g., $=A2*20\%$).

Percentage Increases and Decreases

Excel allows percentages to be written as either a decimal or with a percentage sign (e.g., 0.3 is treated identically to 30%).

- **To calculate a 7% increase:** Multiply the base cell by 1.07 or 107% (e.g., $=B14*1.07$).
- **To calculate a 7% decrease:** Multiply the base cell by 0.93 or 93% (e.g., $=B14*93\%$).
- **Percentage Difference Between Columns:** $=((\text{Current_Value} - \text{Past_Value}) / \text{Past_Value}) * 100$.

Loan Interest Rates (RATE Function)

The financial function $=\text{RATE}(\text{nper}, \text{pmt}, \text{pv})$ determines interest rates based on periodic, constant payments.

- **nper:** Total number of payment periods for the loan.
- **pmt:** The payment made each period (entered as a negative number).
- **pv:** Present value / total loan amount.
- **Note:** If the source data provides the loan duration in *years* rather than periods, you can multiply the year argument by 12 directly inside the formula to find the monthly interest rate (e.g., $=\text{RATE}(C2*12, C3, C4)$). Multiply the final RATE result by 12 to get an annual interest rate.